
Rear USB, USB-C

Ethernet, Video outputs

DIB 2.0

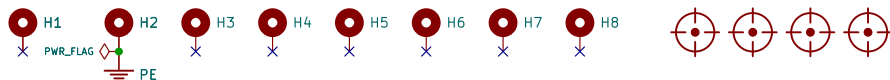
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Power modules coupling

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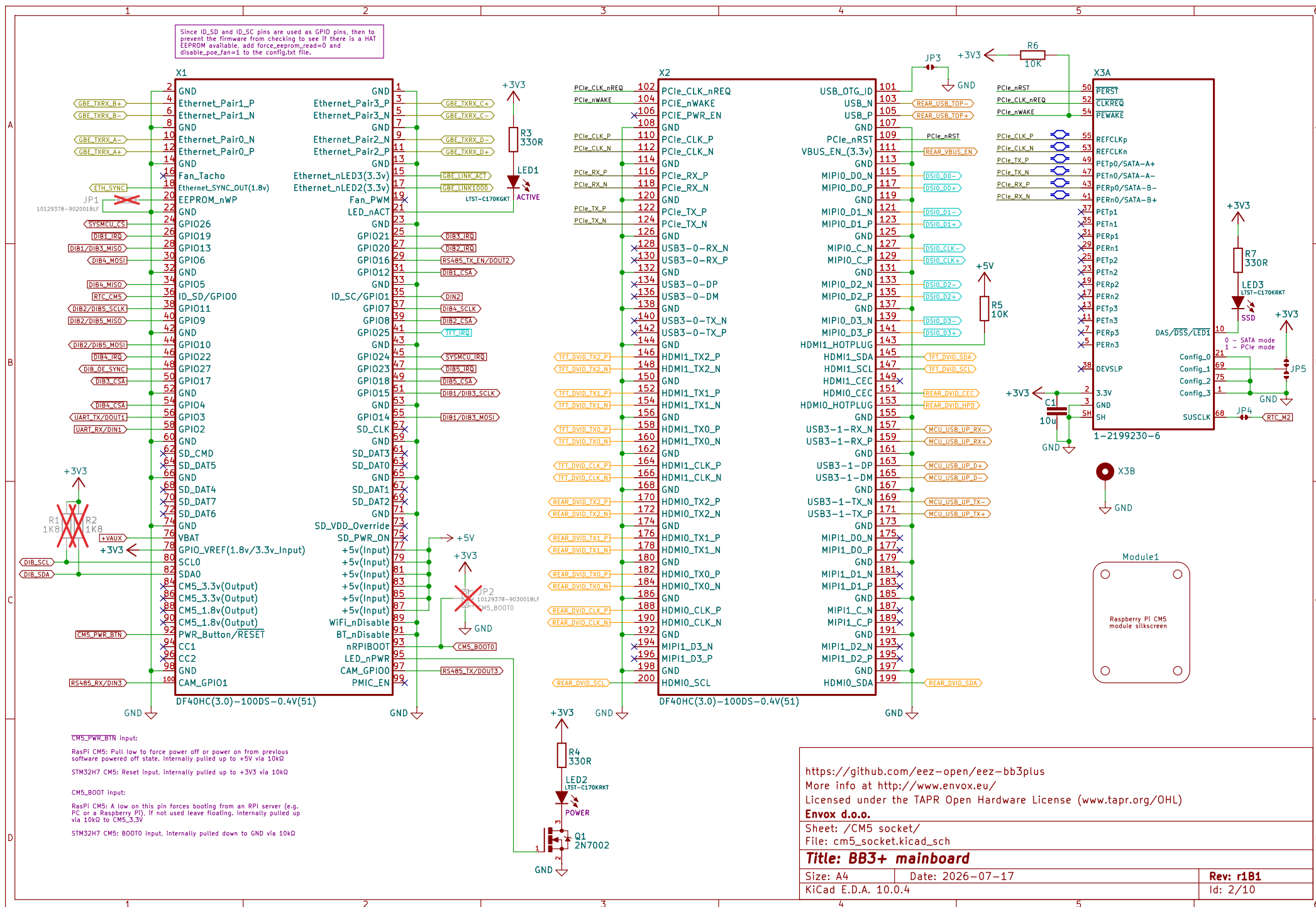
USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	USB FS/H5	USB #0, FS/H5
Rear bottom	USB A	USB Hub P2, H5 only	
Front	USB C	USB Hub P1, H5/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, H5/SS	
DIB #2	DIB 2.0	USB Hub P6, H5/SS	
DIB #3	DIB 2.0	USB Hub P5, H5/SS	
DIB #4	DIB 2.0	USB Hub P4, H5/SS	
DIB #5	DIB 2.0	USB Hub P3, H5/SS	

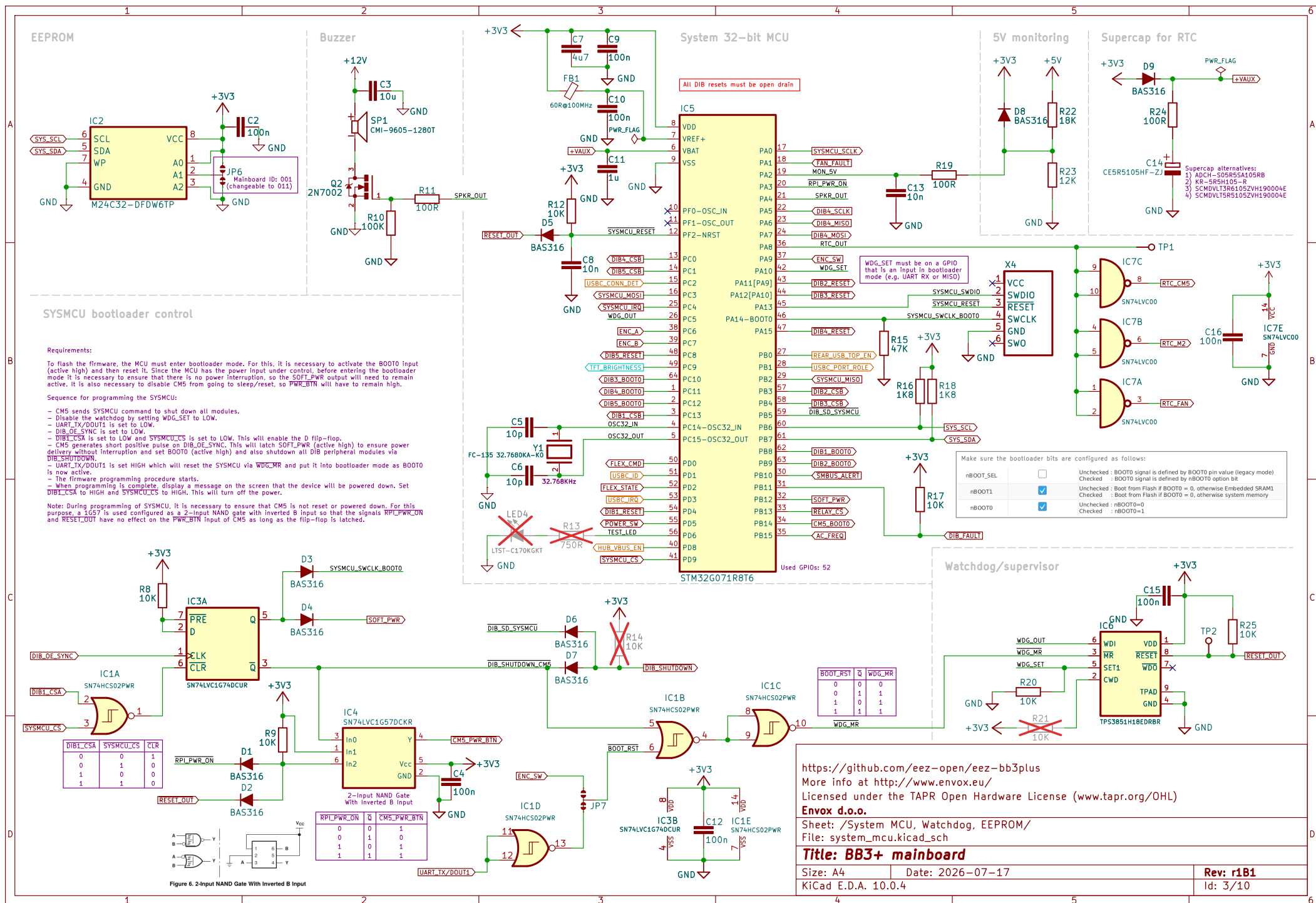
I2C address	Device
0x51	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub?
0x7A	PTN5150A USB CC controller



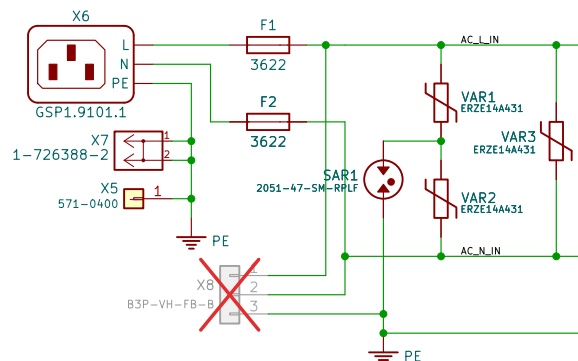
Title: BB3+ mainboard

Id: 1/10

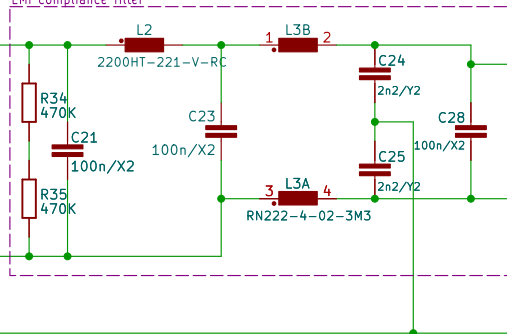




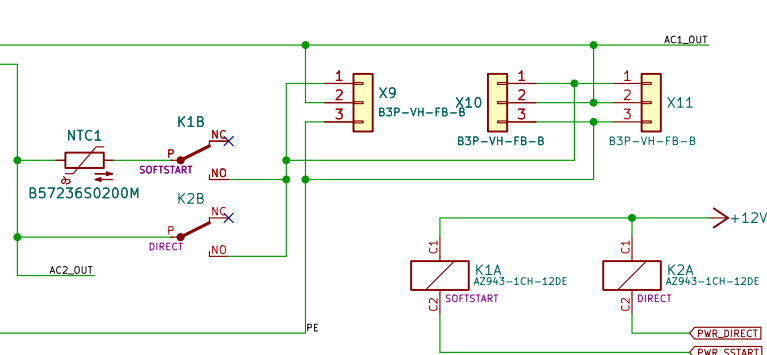
AC inlet, filtering and protection



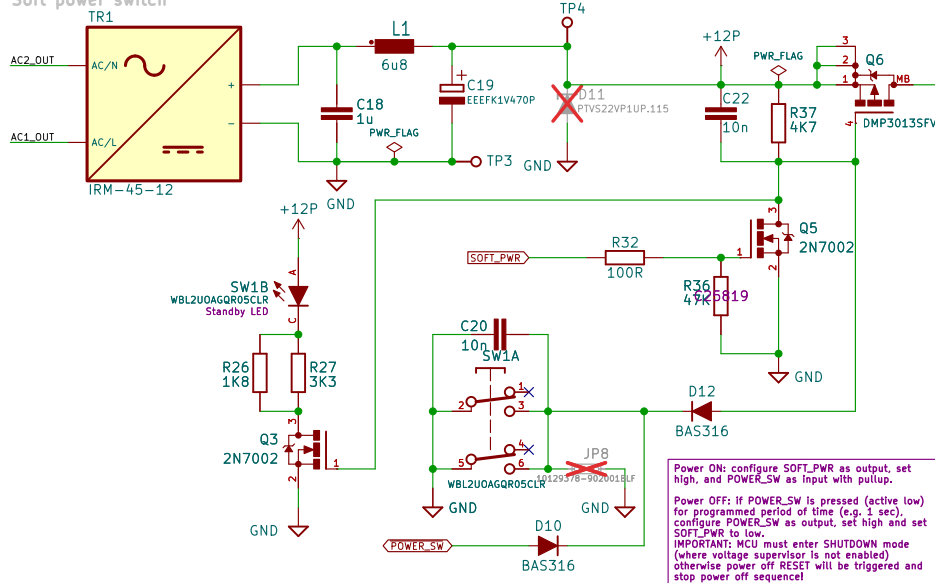
EMI compliance filter



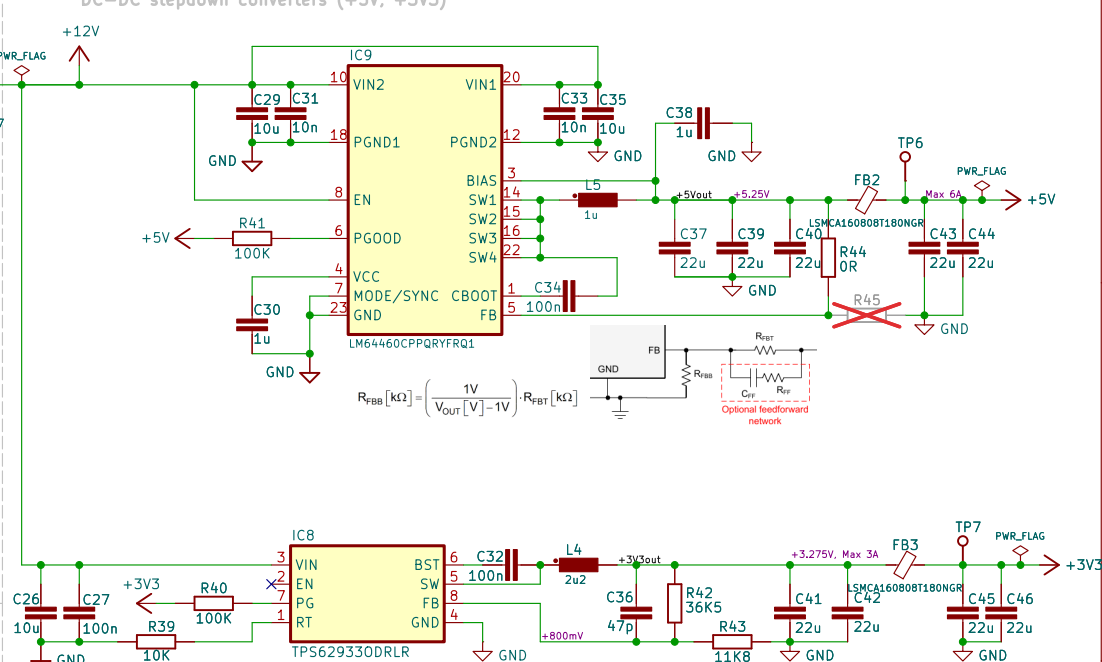
AC power control, Inrush current limiter



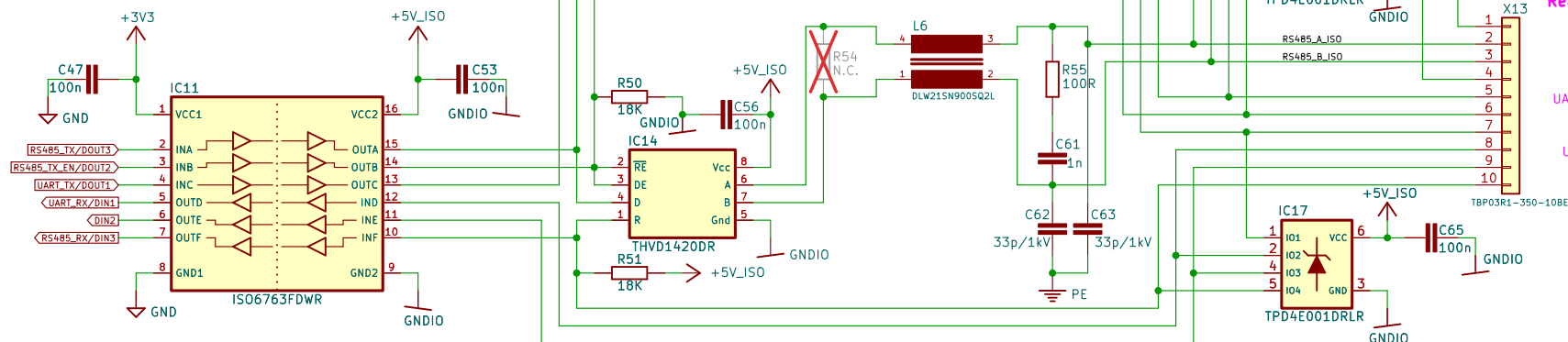
Soft power switch



DC-DC stepdown converters (+5V, +3V3)



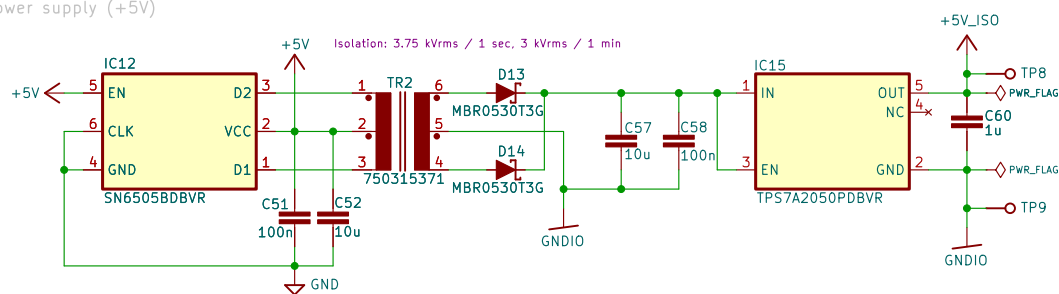
Isolated digital I/O & RS485 transceiver (half-duplex)



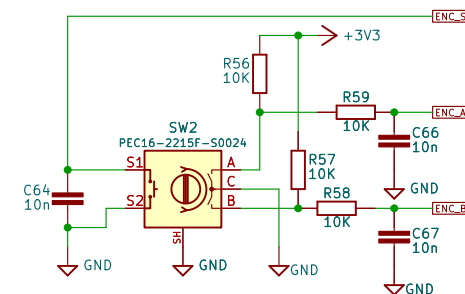
Rear panel I/O connector

- +5V_ISO 1
- RS485_A_ISO 2
- RS485_A_ISO 3
- GND_ISO 4
- UART_TX/DOUT1 5
- DOUT2 6
- DOUT3 7
- UART_RX/DIN1 8
- DIN2 9
- DIN3 10

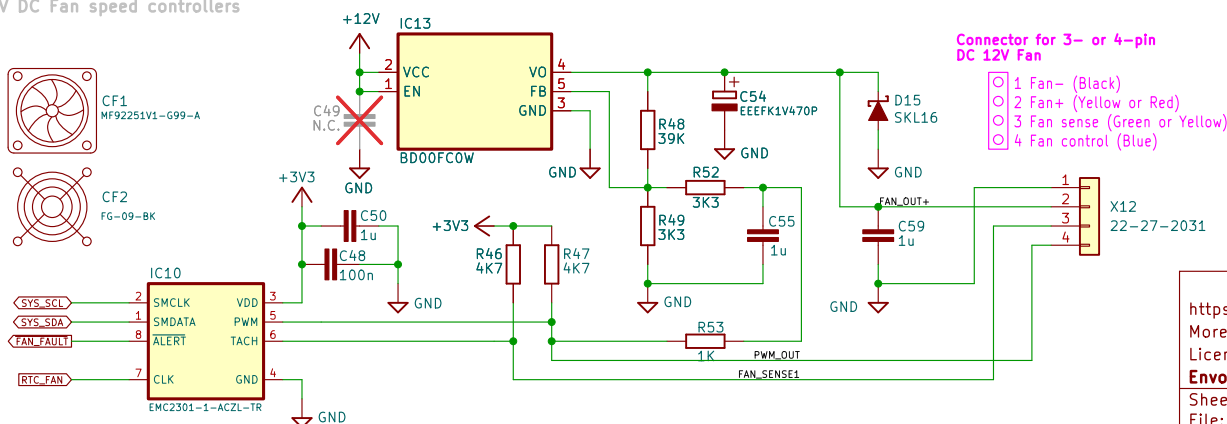
Isolated regulated power supply (+5V)



Encoder with switch



12V DC Fan speed controllers



Connector for 3- or 4-pin DC 12V Fan

- 1 Fan- (Black)
- 2 Fan+ (Yellow or Red)
- 3 Fan sense (Green or Yellow)
- 4 Fan control (Blue)

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

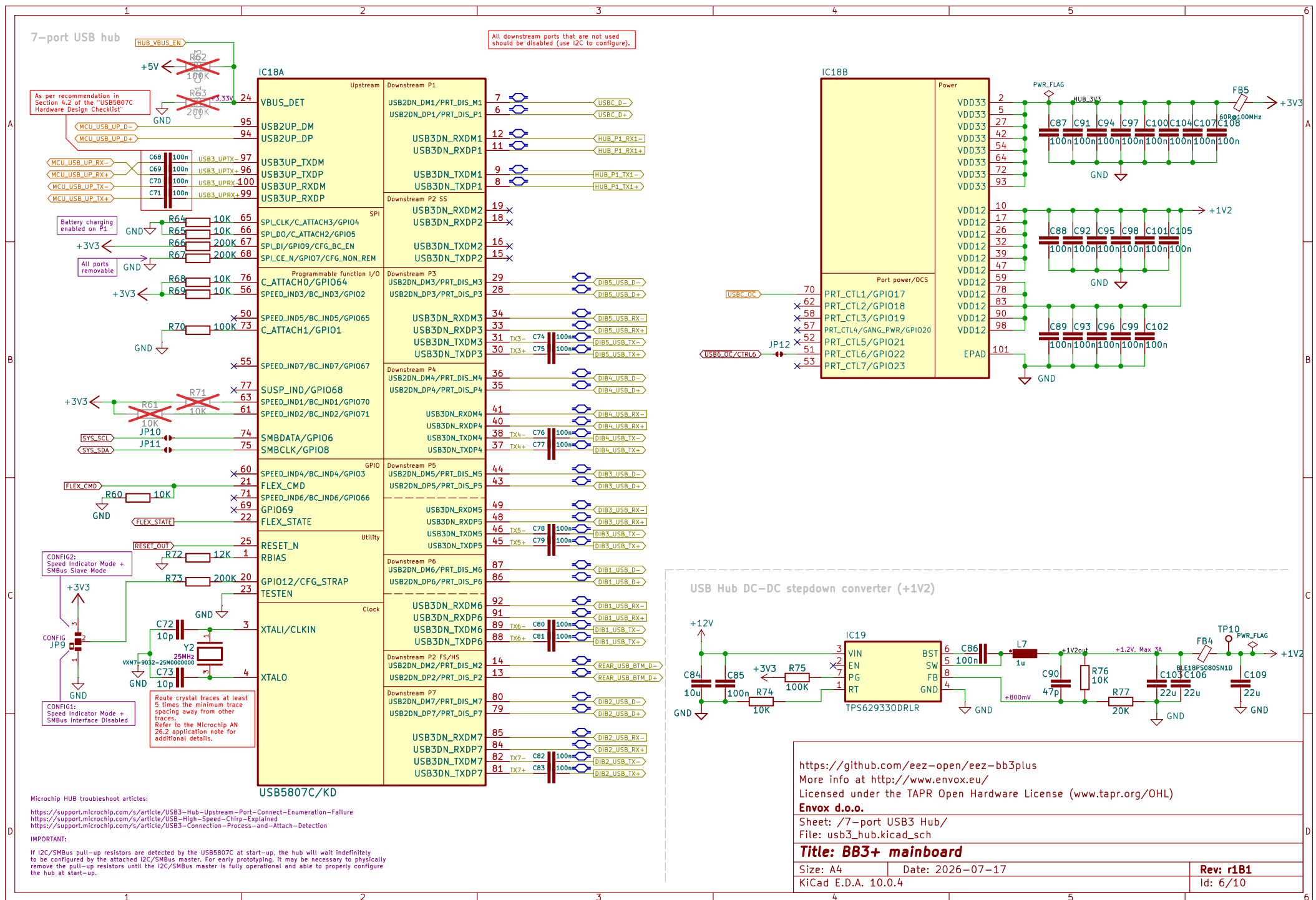
Title: BB3+ mainboard

Size: A4 Date: 2026-07-17

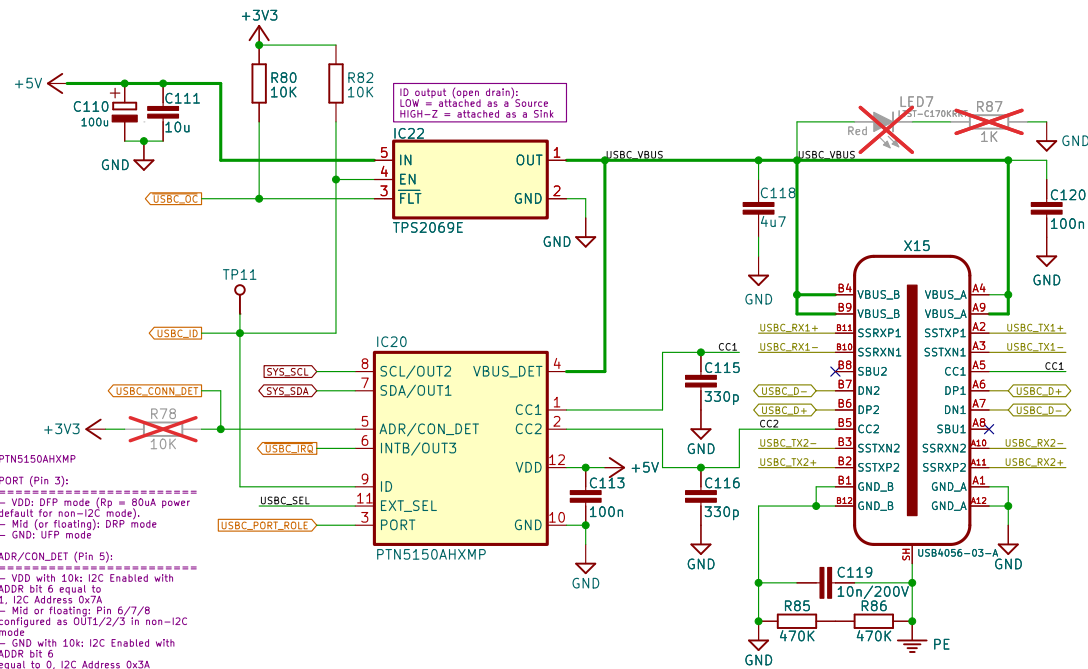
KiCad E.D.A. 10.0.4

Rev: r1B1

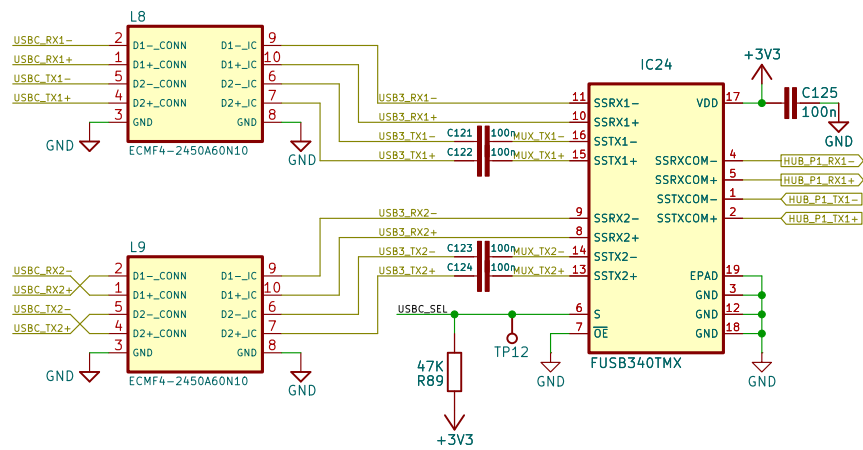
Id: 5/10



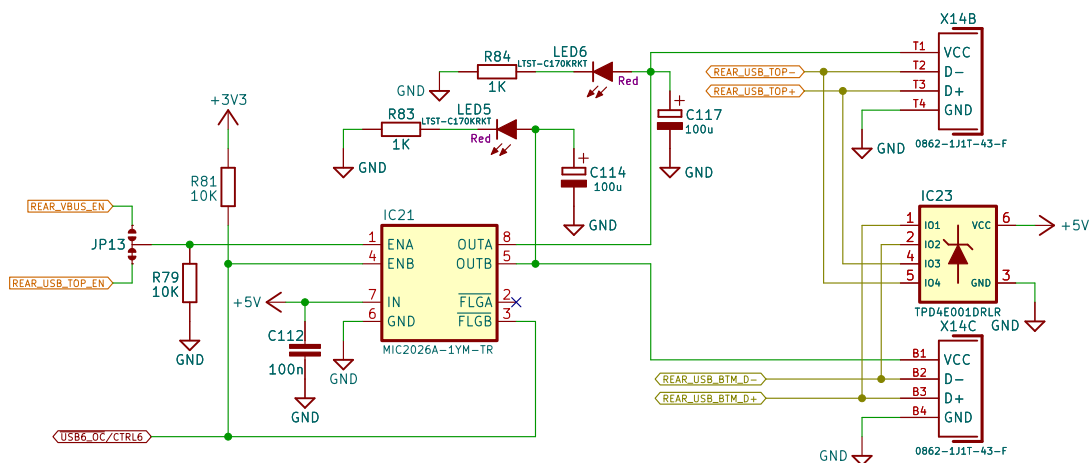
USB-C 3.0 front port (source max 1.5A)



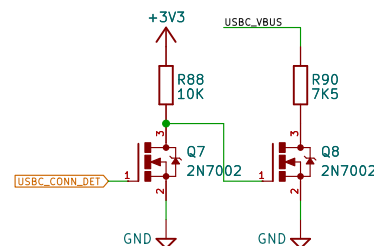
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



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Sheet: /Rear USB, USB-C/

File: usb_conn.kicad_sch

Title: BB3+ mainboard

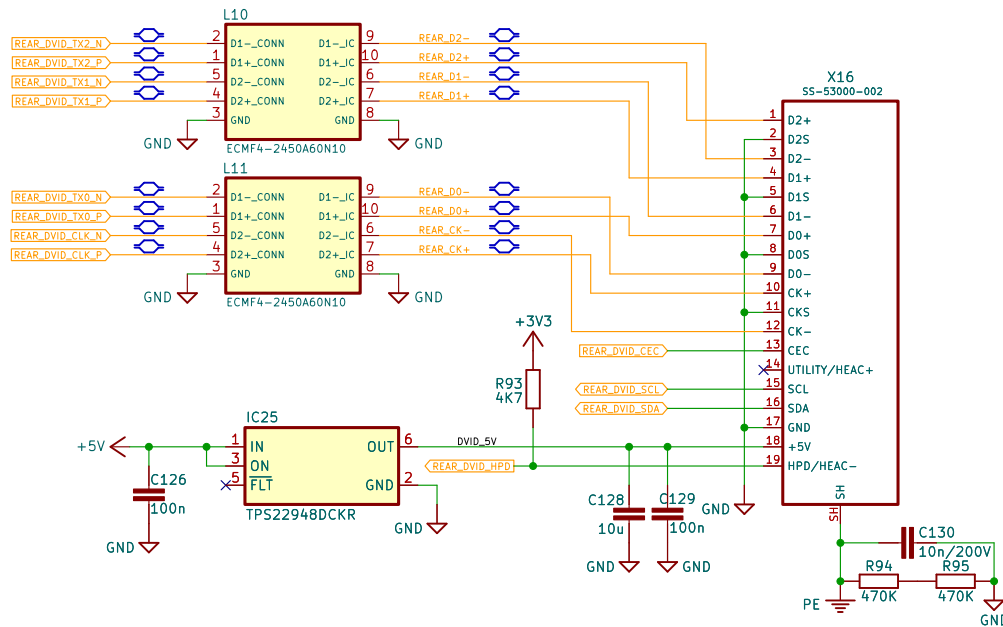
Size: A4 Date: 2026-07-17

KiCad E.D.A. 10.0.4

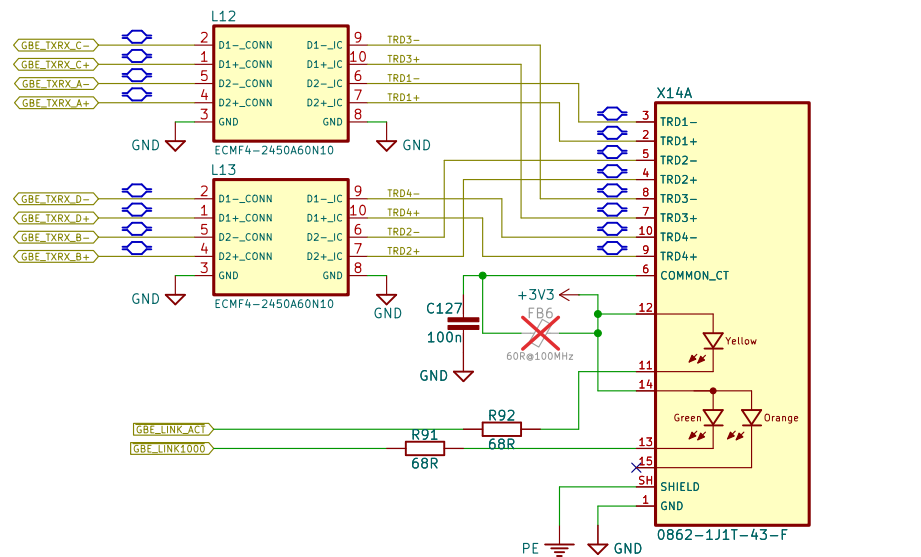
Rev: r1B1

Id: 7/10

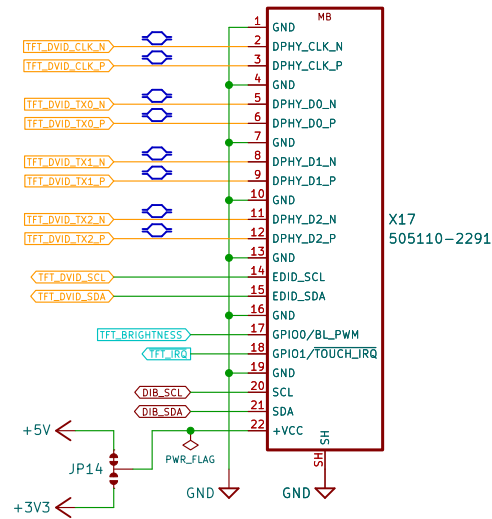
External monitor digital output



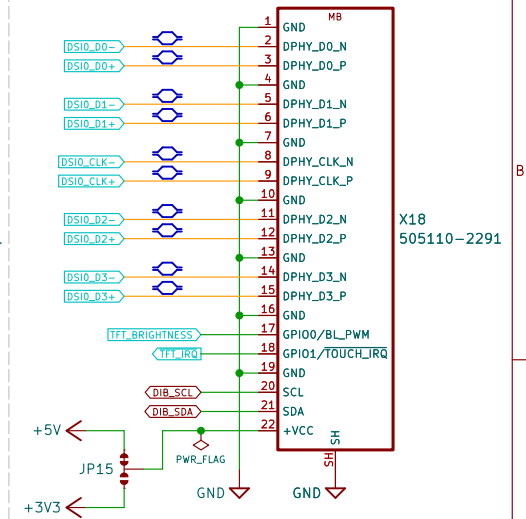
Ethernet RJ45+mag



TFT Touchscreen display DVID output (22-pin FFC)



TFT Touchscreen display DSI output (22-pin FFC)



<https://github.com/eez-open/eez-bb3plus>
 More info at <http://www.envox.eu/>
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Envox d.o.o.

Sheet: /Ethernet, Video outputs/
 File: video_outputs.kicad_sch

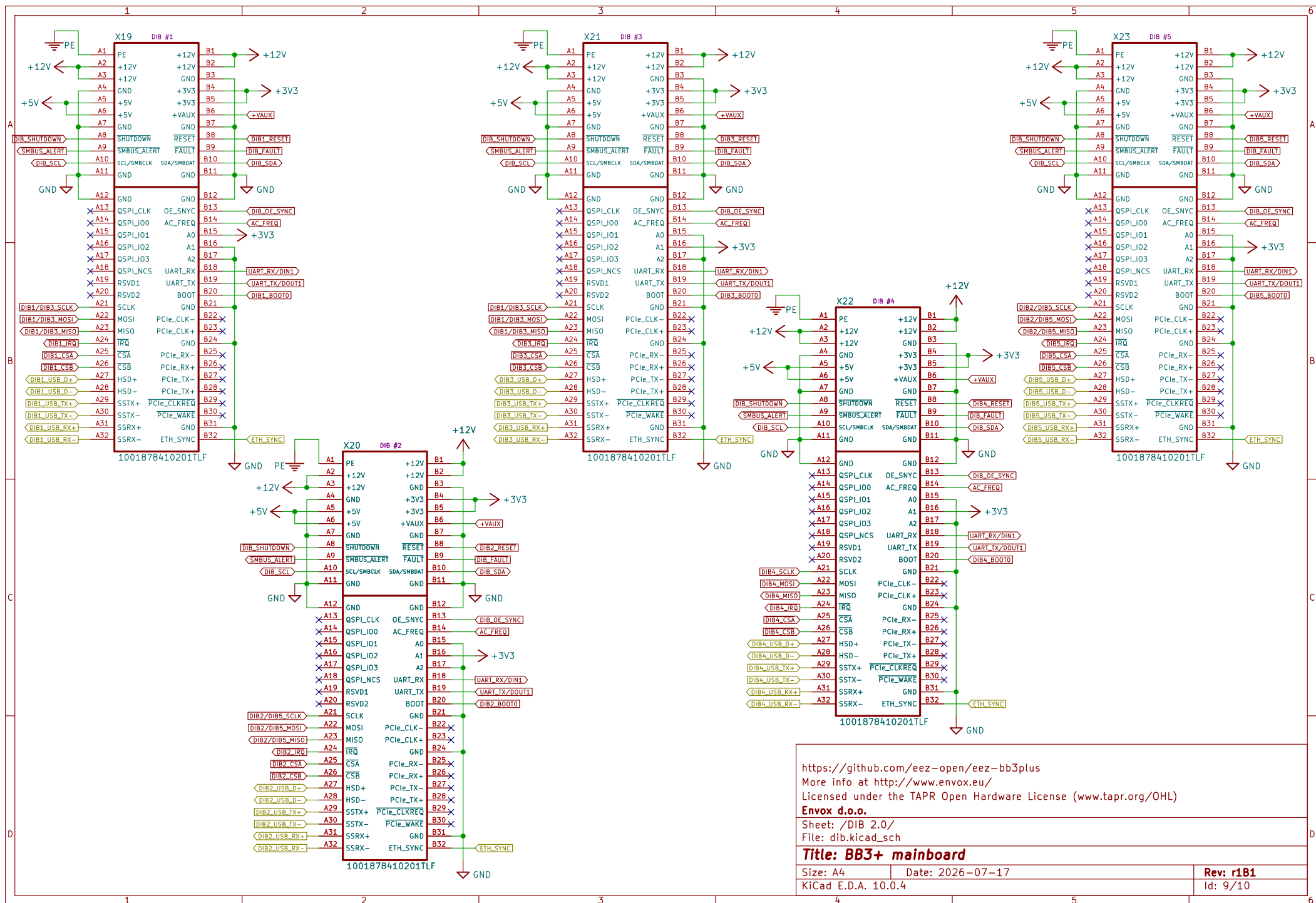
Title: BB3+ mainboard

Size: A4 Date: 2026-07-17

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 8/10



<https://github.com/eez-open/eez-bb3plus>

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Sheet: /DIB 2.0/

File: dib.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-07-17

KiCad E.D.A. 10.0.4

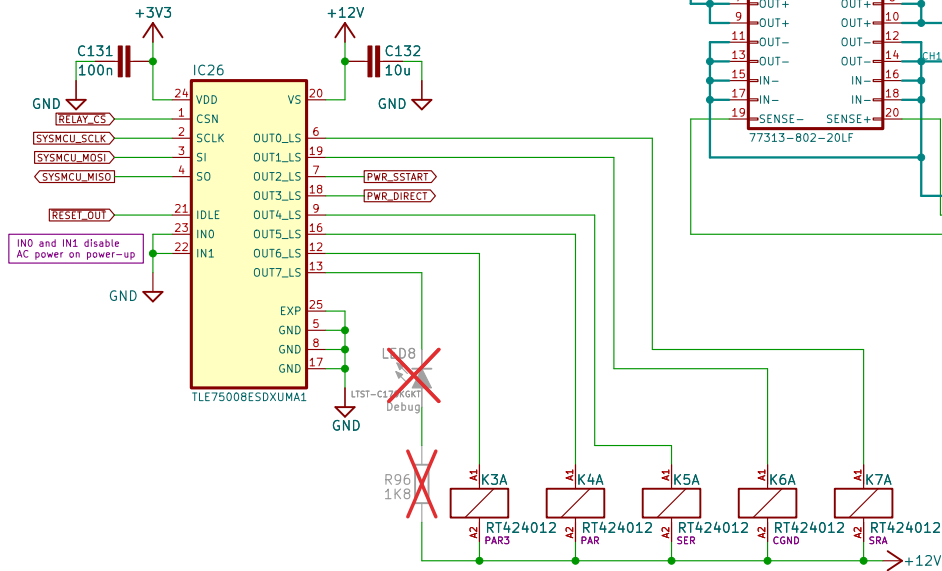
Rev: r1B1

Id: 9/10

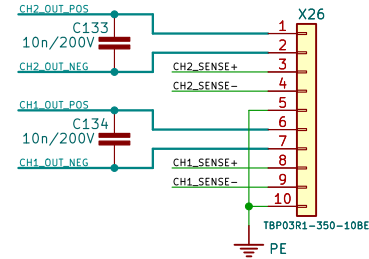
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ IN
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 2	3
CH2_SENSE+ 3	4
CH2_SENSE- 4	5
PE 5	6
CH1_OUT+ 6	7
CH1_OUT- 7	8
CH1_SENSE+ 8	9
CH1_SENSE- 9	10
PE 10	11

<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-07-17

KiCad E.D.A. 10.0.4

Rev: r1B1

Id: 10/10